

Decoding Data Science(DDS) Academy

AI Application Building Challenge

For students (instructor review required)

Day 1 Idea Submission Template

Project Title

Vault – Your money. No excuses

Concept Summary:

(Briefly describe your idea, including its purpose and the problem it aims to solve.)

Vault is a personal finance tracker powered by an AI persona called Frank. Unlike generic finance apps, Vault doesn't just show you where your money went – Frank tells you exactly what he thinks about it, in plain, unfiltered terms.

The core experience: a user uploads their monthly spending spreadsheet (a simple Excel file with dates and category columns works fine), and Vault analyses the data, generates interactive visualisations, and Frank delivers a personalized roast based on the real numbers. Users can also chat directly with Frank to ask questions about their finances.

Because users log in, their data persists, meaning Frank gets more insightful over time, comparing this month to last month, spotting recurring patterns, and tracking whether the user is actually improving.

Target Audience:

(Who will benefit from your project? Describe their needs.)

Primary audience: university students and young adults (18-28) who:

- Struggle to track where their money actually goes
- Find traditional finance apps, limited or overwhelming
- Would engage more with something that has personality and humour
- Want insights without needing to be a finance expert

Secondary audience: anyone who has a simple monthly spending spreadsheet and wants actionable, honest feedback on their habits.

Key Features:

(List the main features or functionalities of your application.)

Authentication & Accounts

- Users sign up and login via Supabase Auth
- All data tied to the user's account — private and persistent
- Session stays active across visits

Excel / CSV Upload

- User uploads their monthly spending file (Excel or CSV)
- Simple format: rows = days/entries, columns = spending categories (food, transport, rent, entertainment, medical, other)
- Parsed instantly using Pandas - handles multiple sheets and missing values

Dashboard & Visualisations

- Total spent this month at a glance
- Spending by category - interactive pie/bar chart (Plotly)
- Spending over time - line chart showing daily/weekly trends
- Month-over-month comparison when previous data exists
- Anomaly highlights - categories that spiked above average

Frank — The AI Roast Persona

- Frank analyses the real spending data - not generic advice
- Delivers a personalised monthly roast: what you overspent on, what patterns he noticed, one genuine piece of advice
- Has a consistent personality: blunt, funny, never sugarcoats
- Visual character - a raccoon mascot (going through your financial trash)

Income, Budget & Savings Tracking

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- User enters income, budget, and net savings (positive or negative) each month alongside their file upload
- Dashboard shows Budget vs Spent comparison and running
- Total Savings to Date
- Gives users a complete monthly financial picture, not just spending

Frank's Monthly Summary

- Frank analyses the month's data - spending, budget, savings and writes a personalised summary
- Blunt, funny tone, references real numbers
- Displayed as a dashboard card, refreshable

Spending History

- Past uploads stored per user (one file per month)
- History page shows month-by-month cards spent, income, budget, savings, verdict badge, top category
- Let's users see their financial trend over time

Technical Approach:

(How do you plan to implement your idea? Mention technologies or methods.)

Technology	Purpose
Python	Core language for all logic
Streamlit	Frontend UI and multipage app structure
Pandas	Excel/CSV parsing, data analysis, pattern detection
Plotly	Interactive carts – pie, bar, line graphs
Groq API	Power's Frank's AI persona and chat responses
Supabase	User authentication + persistent data storage
Streamlit Community Cloud	Free cloud deployment for the final app

Expected Challenges:

(What potential obstacles do you foresee, and how will you address them?)

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- Supabase authentication setup
- Varied Excel formats from users
- Streamlit UI limitations
- Frank's roasts feeling generic
- Running out of time to complete all features

Submission Format:

(Outline how your submission should be presented: documents, slides, etc.)

- GitHub repository with a clean README explaining the app, tech stack, and how to run it locally
- Deployed live link on Streamlit Community Cloud
- Short demo video – showing the upload flow, dashboard, Frank's roast, and chat page
- Final LinkedIn post documenting the build journey and final product
- Progress workbook
- Presentation slide deck talking of the project

Expected Outcome:

(What do you aim to achieve by the end of the challenge?)

A fully deployed, working finance web app with:

- Working user authentication and persistent accounts
- Excel uploads and real data analysis using Pandas
- Interactive Plotly charts showing spending patterns
- A named AI persona (Frank) that roasts user's based on real data (their data)
- Running savings total and showing financial progress over time

Additional Notes (Optional):

- The raccoon mascot (Frank) should be consistent across all pages, a small avatar in the nav, a larger illustration on the dashboard roast section, and a chat avatar.
- Provide a downloadable sample Excel template in the app so users know exactly what format to upload
- Keep the core loop (upload → charts → roast) as the absolute priority. Chat and history are stretch goals
- After the challenge, the dream version of Vault is rebuilt with Flask + React + Supabase for a proper portfolio centrepiece

The name Vault is a play on 'receipts' (evidence/proof) and 'wrecked' - memorable, on-brand, and self-explanatory

- Chat with Frank feature was moved to v2 to keep the 8-day scope focused on a complete, polished core experience